A pre-analysis diagnostic for reference-cohort model-X knockoffs in neuroimaging

Feature geometry sets the ceiling; the number of true signals, cohort mismatch and the penalty constant govern the outcome

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Abstract

Model-X knockoffs give finite-sample FDR control under an arbitrary response model, but in neuroimaging the covariate distribution must be estimated from a reference cohort, and the procedure often reports nothing. We supply a diagnostic computable from the reference cohort before any knockoff is drawn, and establish what it predicts.

The diagnostic is the effective rank of the ROI summary's covariance, set by the summary, not the cohort. Averaging Fisher-z connectivity over all other parcels—the field's default—makes each feature a row mean of a correlation matrix, hence a proxy for its first principal component: median $R^2 = 0.677$, effective rank 2.56 of 113. Partial correlation gives 71.6, unregularised 59.8. The four summaries agree at median $r \leq 0.26$, every ROI below 0.5.

Known-truth controls delimit this. A low effective rank does not explain a null result: with the real covariance the pipeline attains power 0.859; reference-cohort size buys a few points of power, not the outcome; what binds is the number of true signals (0.97 at $k = 12$, 0.20 at $k = 3$). Six constructions are indistinguishable; a mirror method keeps validity only by giving up power.

Validity is separate. Cohort mismatch of 0.12–0.19 breaks FDP control while *raising* power; after subtracting a sampling-noise floor it is negligible in ABIDE I and II but 0.117 in ADHD-200, whose FDP is exceeded. The penalty constant must satisfy power and convergence at once.

A cohort of another disorder replicates the ordering but not the severity: the diagnostic generalises, the constants do not.

Keywords: model-X knockoffs · false discovery rate · reference cohort · neuroimaging · effective rank · reproducibility

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1 Introduction

The problem. Identifying which of many measured quantities a phenotype depends on is central to neuroimaging. A region-of-interest (ROI) study measures $p \sim 10^2$ features per subject and asks which are associated with a diagnosis or a cognitive score. Testing each feature marginally inflates false discoveries, and the conventional remedy—Benjamini–Hochberg on p-values—presupposes valid p-values that, for the models practitioners fit, either do not exist or are badly non-uniform in finite samples (Dezeure et al., 2015). The arithmetic is unforgiving: brain-wide association effect sizes are $r \approx 0.01$–0.05, so reproducible individual-level inference requires thousands of subjects (Marek et al., 2022), and cluster-based fMRI inference fails badly enough to have been documented in detail (Eklund et al., 2016).

Model-X knockoffs. The knockoff filter (Barber and Candès, 2015) sidesteps p-values. Its original form assumes a homoscedastic linear model with $n \geq 2p$; the model-X extension (Candès et al., 2018) removes both restrictions. By treating the covariates as random and assuming F_X *known*, it gives exact finite-sample FDR control with no assumption on the response model and no constraint on p relative to n. Construct a synthetic copy $\tilde{\mathbf{X}}$ of the covariates satisfying pairwise exchangeability and independent of the response given $\mathbf{X}$; under the null the sign of the knockoff statistic is an independent fair coin flip, so the number of large *negative* statistics estimates the number of false discoveries among the large positive ones. The framework has been generalised to more efficient samplers (Bates et al., 2021), faster conditional randomisation (Liu et al., 2022), better constructions (Spector and Janson, 2022), and sharper power theory (Wang and Janson, 2022; Liu and Rigollet, 2019; Ke et al., 2024).

The price is F_X, and the received explanation for the bill. F_X being known is the framework's entire cost, and in neuroimaging it cannot be met: the covariate distribution must be estimated from an independent reference cohort. That this substitution is not benign is known on the validity side—FDR inflation is bounded by a divergence between true and working conditional distributions (Barber et al., 2020), exact control survives estimated parametric families with $\Omega(n^* p)$ parameters (Huang and Janson, 2020), and asymptotic control holds under coupling (Fan et al., 2025), moment matching (Fan et al., 2026) or data splitting (Fan et al., 2020). It is also known on the practical side, in the adjacent literature on genome-wide association, where knockoffs and fine-mapping methods estimate linkage disequilibrium from an external reference panel and where panel mismatch is a documented source of FDR inflation (Benner et al., 2017; Chen et al., 2021; Kanai et al., 2022).

Knockoffs are now applied to ROI-level connectivity and to network comparisons between groups (Ji et al., 2024), and alternatives to the equicorrelated construction are benchmarked on exactly this kind of data (Chen et al., 2023). The explanation we examine here is the first one those applications invite: the reference covariance is nearly singular, the equicorrelated construction reads $\lambda_{\min}(\hat{\Sigma})$ directly, and the resulting knockoffs are too close to the originals to be distinguished

from them. That description of the spectrum is correct, and the construction literature has addressed its consequences (Spector and Janson, 2022; Candès et al., 2018). Our question is whether it explains the outcome.

What this paper does. We supply a diagnostic that a practitioner can compute from the reference cohort alone, before any knockoff is drawn, and we establish what it predicts and what it does not. The diagnostic is the effective rank of the ROI summary's covariance. It predicts power; it does not predict validity, and we give the two conditions that separately govern the latter. Arriving at it required first testing the explanation the field reaches for—that the reference covariance is too degenerate to build knockoffs from—and finding that it fails; we report that rejection because it is the reason the diagnostic points where it does. Figure 1 summarises the design and fixes the scope of the claim. The argument has three moves.

First, the hypothesis fails its own test. Three independent manipulations of the covariance—using the real covariance as ground truth, varying reference-cohort size from 100 subjects to none, and introducing the distribution shift actually observed between ABIDE's cohorts—each fail to reproduce the null result (§3.2–3.3). What limits power in simulation is the number of true signals, a property of knockoff+ itself.

Second, we exhaust the machinery. We compare six knockoff constructions, a mirror-based method that never inverts Σ, a multi-environment filter that tests cross-site consistency, and a one-parameter family of matrix-power transforms. None of them rescues the outcome (§3.4–3.5), and we report for each what the evidence actually supports—including the multi-environment filter, which turns out to have too little power in this configuration to support any conclusion, a fact its own positive control establishes. The matrix-power family is worth singling out: it produces a continuous curve along which the procedure's own reported error improves monotonically while its true error degrades monotonically, and a global-null control confirms the crossing.

Third, the feature geometry sets the ceiling. Averaging Fisher-z connectivity over all other parcels—the field's default, global brain connectivity (Cole et al., 2012)—makes each feature a row mean of a correlation matrix, and a row mean is a proxy for the first principal component. In our data the median R^2 of a single feature on PC1 is 0.677 and the effective rank is 2.56 of 113. That the row sum of a connectivity matrix is dominated by PC1 has been observed at the group level (van den Heuvel et al., 2025) and for mean connectivity maps (Moujaes et al., 2024); we quantify it per subject, per feature, in a variable-selection context, and show that it bounds what any downstream method can achieve (§3.1). It does not, however, set the power: across the 41 covariance configurations we sweep, spanning effective rank 2.2 to 113, per-signal power never falls below 0.770, whereas changing the number of true signals moves it from 0.97 to 0.20.

We state at the outset what this paper is not. It is not a claim that model-X knockoffs are invalid, nor that they cannot work in imaging. The method is used here as designed, and it is the only framework in this literature with a finite-sample guarantee to characterise. Our contribution is to locate where that guarantee is achievable and where it is not, and to supply a diagnostic—computable before any analysis is run, from the reference cohort alone—that tells a

practitioner which side of the boundary they are on.

2 Methods

2.1 Data

ABIDE I (Di Martino et al., 2014) was obtained from the FCP-INDI public archive in the `cpac/filt_noglobal` branch (1,035 subjects, 20 sites), together with the `niak/filt_noglobal` branch for the pipeline comparison of §3.10. Subjects with `func_mean_fd` ≥ 0.2 were excluded (1,035 $\rightarrow$ 885; 14%).

We additionally observed *constant* parcel time series—a signature of incomplete brain coverage—in 68 of 885 subjects (7.7%), concentrated by site (CMU 80%, SBL 69%, CALTECH 46%, PITT 29%; all others $\leq 5\%$) and by region (60% of affected parcels in the inferior cerebellum and vermis). Such channels produce undefined correlations; we mark them missing rather than imputing a value, remove parcels missing in > 20 subjects, then remove subjects incomplete on the retained parcels. For AAL116 (Tzourio-Mazoyer et al., 2002) this yields $p = 113$, $n = 852$; the procedure was applied identically to Harvard–Oxford and CC200 (Craddock et al., 2012).

ABIDE II (Di Martino et al., 2017) was obtained from the same archive (`Derivatives/DCAN`; Power2011 parcellation, CIFTI parcel time series, $p = 94$, $n = 956$). The two cohorts share the disorder and almost nothing else: different participants, scanners, sites, parcellation, pipeline and file format.

ADHD-200 (Bellec et al., 2017) is the third cohort and the out-of-sample test of §3.6 and §3.11. It differs from both ABIDE cohorts in disorder, participants, scanners, sites, preprocessing pipeline and parcellation. We used the Athena-preprocessed release with the AAL116 parcellation, retaining only runs whose distributed quality flag is `1`: of 768 subjects, 93 failed and 11 at the three Peking sites are flagged as unevaluated, which we excluded conservatively, leaving $n = 664$ (427 controls, 237 ADHD, nine sites; age 7.1–21.8). Quality-aware run selection means a subject whose first resting run failed was kept if another run passed, rather than dropped whole. No retained subject has a constant parcel time series, and the feature matrix has no missing entries, so the missingness handling of the ABIDE I paragraph above does not apply here. Head-motion records are distributed for one site only (KKI, $n = 78$; peak displacement 0.26–3.91 mm, median 1.09), so no motion threshold was applied to this cohort on the ABIDE I criterion, and we state this as a limitation rather than a matching of pipelines.

Two properties of this cohort matter for how §3.11 can be read. First, the AAL116 labelling here was verified to enumerate the same parcels in the same order as the ABIDE I feature matrices, so feature j denotes the same region in either cohort and the two can be compared directly. Second, Athena is not CPAC or NIAK, so this is a cross-pipeline comparison of the kind §3.10 treats as coarser than the summary choice; we therefore use ADHD-200 to test the *mechanism*—whether a summary's effective rank and the cohort-shift boundary behave the way §3.1 and §3.6 predict—and not to expect region-level agreement.

Main/reference split. All three cohorts are split the same way throughout: the main cohort is the six largest sites and the reference cohort is every remaining subject, so that the reference distribution is estimated from subjects the main analysis never sees. For ABIDE I this gives $n_{\text{main}} = 459$ and $n_{\text{ref}} = 393$; for ABIDE II, 564 and 392; for ADHD-200, 543 and 121. The resulting n_{ref}/p is 3.5, 4.2 and 1.0 respectively, which is why ADHD-200 is only ever compared at matched reference size (§3.11): with $n_{\text{ref}} \approx p$ the sample correlation matrix is nearly singular by construction, and reading that as a property of the cohort would be an artefact.

2.2 ROI summary definitions

Let r_{ikj} be the Pearson correlation between parcels j and k in subject i, over valid parcels only. We consider four summaries, each producing one scalar per parcel:

$$\text{meanconn:} \quad x_{ij} = \frac{1}{n_{\text{valid}} - 1} \sum_{k \neq j} \operatorname{arctanh}(r_{ikj}) \tag{1}$$

$$\text{gsr:} \quad \text{global-signal regression, then (1)} \tag{2}$$

$$\text{pcresid:} \quad \text{remove the top 5 principal components, then (1)} \tag{3}$$

$$\text{partial:} \quad \text{as (1) but with partial instead of Pearson correlations} \tag{4}$$

Definition (1) is the field's default: it is global brain connectivity, or weighted degree centrality, introduced as a per-region individual-difference feature by Cole et al. (2012). It is a *row mean of a correlation matrix*. Definitions (2) and (3) are the two standard neuroimaging remedies for the global-signal component; (4) targets the quantity knockoffs test, namely conditional rather than marginal association.

Partial correlation requires the inverse of a $p \times p$ correlation matrix, which does not exist when the number of volumes T does not exceed p. This is not hypothetical: 131/885 CPAC and 159/885 NIAK subjects have $T \leq n_{\text{valid}}$ (shortest $T = 41$). We therefore estimate the precision matrix by Ledoit–Wolf shrinkage (Ledoit and Wolf, 2004). The choice is not load-bearing for our conclusions: §3.1 reports the effective rank of the resulting features across shrinkage strengths from 0 to 0.5, including the unregularised case, and it varies only between 53 and 67.

2.3 Reference-cohort knockoffs

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the main-cohort covariates and y the response. We use the Gaussian equicorrelated construction, $s_j \equiv 2\lambda_{\min}(\hat{\Sigma}) \wedge 1$, unless stated otherwise, and knockoff+

$$\tau_+ = \min\left\{t > 0 : \frac{1 + \#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\}} \leq q\right\}, \qquad \hat{S} = \{j : W_j \geq \tau_+\}, \tag{5}$$

which controls FDR $\leq q$ in finite samples. $\hat{\Sigma}$ is estimated from the reference cohort by Ledoit–Wolf shrinkage; the reference cohort's responses are never used.

The 20 ABIDE I sites are partitioned by size: the six largest (NYU, UM_1, USM, UCLA_1, YALE, PITT) form the **main cohort** ($n_{\text{main}} = 459$; ASD/control 218/241); the remaining 14

form the **reference cohort** ($n_{\text{ref}} = 393$; 174/219). Because both pass through an identical pipeline and parcellation, features are exactly commensurate, while differing scanners and participant composition make the distributional mismatch real rather than simulated. ABIDE II is split the same way ($n_{\text{main}} = 564$, $n_{\text{ref}} = 392$).

2.4 The feature statistic, and its penalty constant

We use a fixed-λ lasso coefficient-difference (LCD) statistic,

$$W_j = |\hat{b}_j| - |\hat{b}_{j+p}|, \qquad \hat{b} = \arg\min_b \ \tfrac{1}{2}\|y - [\mathbf{X}, \tilde{\mathbf{X}}]b\|_2^2 + \lambda\|b\|_1, \tag{6}$$

with $\lambda = c \cdot \max_k |([\mathbf{X}, \tilde{\mathbf{X}}]^\top y)_k|/n$. Because swapping a column with its knockoff leaves the *set* of columns unchanged, λ is invariant to swapping, so (6) satisfies the flip-sign property. We avoid cross-validated tuning because it is not reproducible on this data (§3.9).

The constant c is not a free parameter in the sense usually assumed. It must satisfy two constraints simultaneously, and they pull in opposite directions: small enough that an injected signal survives the penalty, and large enough that coordinate descent converges on the real, weak response (§3.7). Earlier versions of this work used $c = 0.15$, which violates the first. Every result below uses $c = 0.05$ and reports the fraction of replications in which the solver converged; rows below 0.90 are excluded from the tables and marked where retained.

2.5 Spectral diagnostics

For a correlation matrix C with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$ we report $\lambda_{\min}$, the condition number, the first-component share $\lambda_1 / \sum_i \lambda_i$, and the effective rank $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ (Roy and Vetterli, 2007). These are computable from the reference cohort alone, before any knockoff is drawn.

2.6 Controls: why injection and global nulls carry the argument

On real data there is no ground truth for which ROIs are genuinely associated with a diagnosis. We therefore cannot compute a false discovery proportion on the real response; the only error number available there is the procedure's own report, $(1 + F_\tau)/R_\tau$, which is exactly what is under examination. Every quantitative claim about error in this paper comes from one of two constructions in which the truth is known by construction:

- **Injection.** A response $y = \mathbf{X}\beta + \varepsilon$ is generated with β supported on k known coordinates. The remaining $p - k$ are null by construction, so FDP is exact.
- **Global null.** The response is drawn independently of $\mathbf{X}$, so every variable is null and every discovery is false by construction. Then $\mathbb{E}[\text{FDP}] = P(\hat{S} \neq \emptyset)$, and knockoff+ guarantees this is $\leq q$; comparing the observed rate against q is a test of validity that requires no biological ground truth.

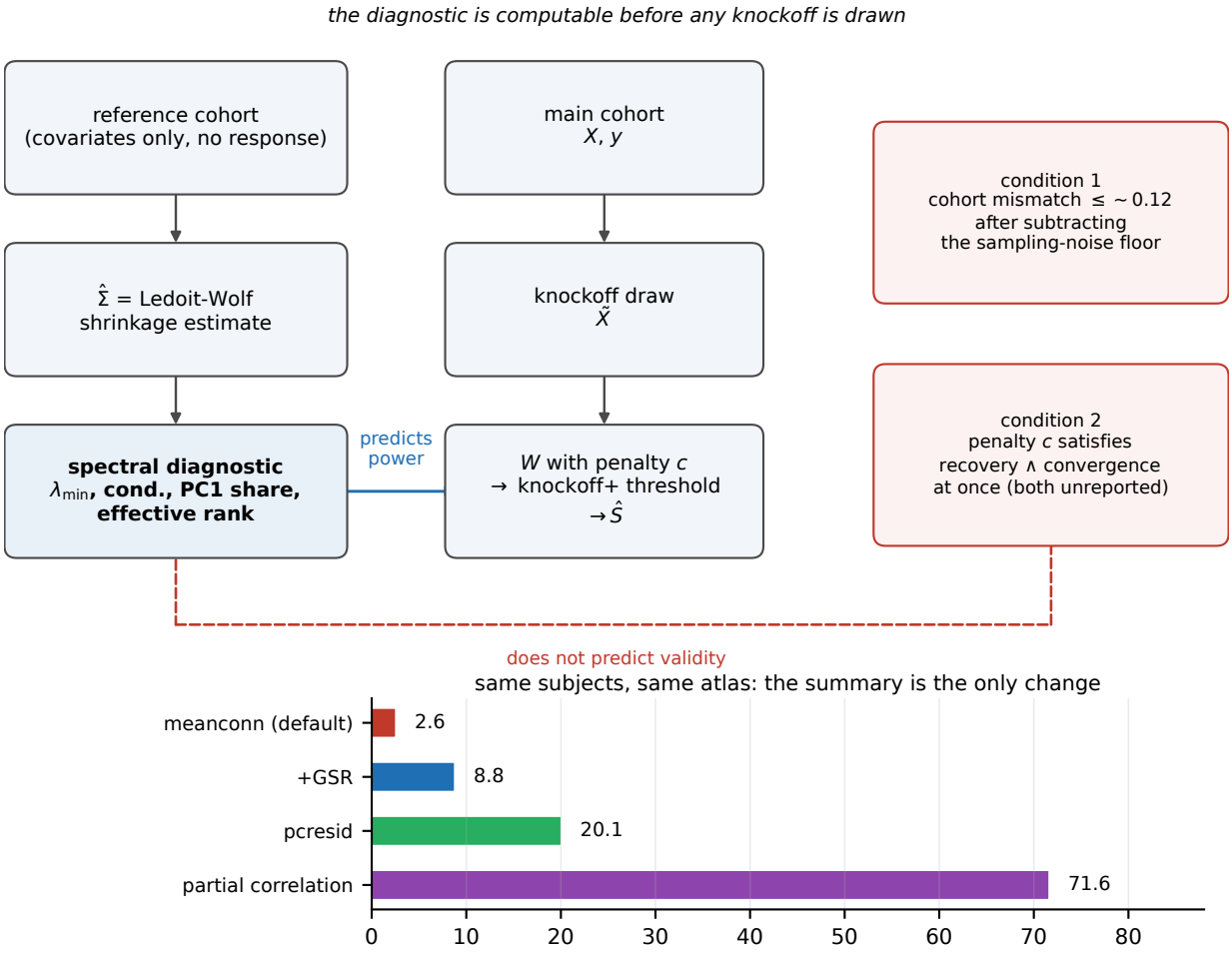


Figure 1: The diagnostic is computable before any knockoff is drawn, and it answers a narrower question than a user may expect. Left column: the reference cohort, which carries covariates only, yields $\hat{\Sigma}$ and the four spectral quantities of §2.5. Centre column: the main cohort and the knockoff draw. Right column: the two conditions that govern *validity*, neither of which the diagnostic predicts. The diagnostic predicts power (solid arrow); it does not predict validity (dashed arrow), which depends on cohort mismatch and on the penalty constant satisfying two conditions at once. Bottom: the same subjects and the same atlas under four summary definitions, which is the quantity the rest of the paper turns on.

Injection amplitudes are calibrated on the *conditional* $|t|$ of the injected coordinates, which is the quantity knockoffs test. This matters: for the ROIs with the highest variance-inflation factors, a target conditional $|t| = 6$ corresponds to a marginal $|t| = 48.9$, an inflation of 8.2×, rising to 13.0× for random supports at low signal. Calibrating on the marginal $|t|$ injects a contrast that is almost entirely contained in the span of the remaining columns and measures nothing about the procedure.

All experiments use $q = 0.20$, $c = 0.05$ and ≥ 60 replications (typically 200). Result tables, provenance records and a script that maps each numeric claim in this manuscript to a specific cell of a specific result file are archived as described in the Declarations.

3 Results

3.1 The ROI summary determines the geometry of the feature space

Table 1 reports the spectrum of the reference cohort's correlation matrix for four summary definitions across five dataset/parcellation combinations, and Figure 2 displays the same quantities.

		effective rank of the reference covariance			
dataset	p	meanconn	+GSR	pcresid	partial
ABIDE I, AAL116, CPAC	113	2.56	8.79	20.06	**71.63**
ABIDE I, AAL116, NIAK	116	3.23	3.31	17.97	**72.66**
ABIDE I, CC200, CPAC	200	2.38	14.71	27.50	**105.16**
ABIDE I, Harvard–Oxford	109	2.23	12.01	22.12	**72.88**
ABIDE II, Power2011	94	15.95	9.67	26.36	23.92

Table 1: Effective rank $(\sum \lambda_i)^2 / \sum \lambda_i^2$ of the reference cohort's correlation matrix, by ROI summary definition. The conventional definition sits at the bottom of the range in every ABIDE I configuration.

The mechanism is elementary and worth stating, because it is what makes the result a property of the definition rather than of functional connectomes. Feature (1) is a row mean of a correlation matrix. In a correlation matrix whose leading eigenvalue is large, the row mean of row j is dominated by the j-th entry of the leading eigenvector, and the vector of row means is therefore a proxy for that eigenvector. In our data the first component carries 62% of the total variance, the effective rank is 2.56 of 113, and the median R^2 of a single feature on PC1 alone is 0.677, against 0.288, 0.067 and 0.018 for GSR, pcresid and partial correlation respectively (Figure 3).

That the row sum of a connectivity matrix is dominated by its first principal component is not new as a general observation: it has been established at the group level for normative connectivity matrices (van den Heuvel et al., 2025), where the first three components exceed 70% of the variance and degree correlates with PC1 at $|r| = 0.82$, and for mean-connectivity maps, where the correlation is $r = 0.86$ without global-signal regression (Moujaes et al., 2024). What

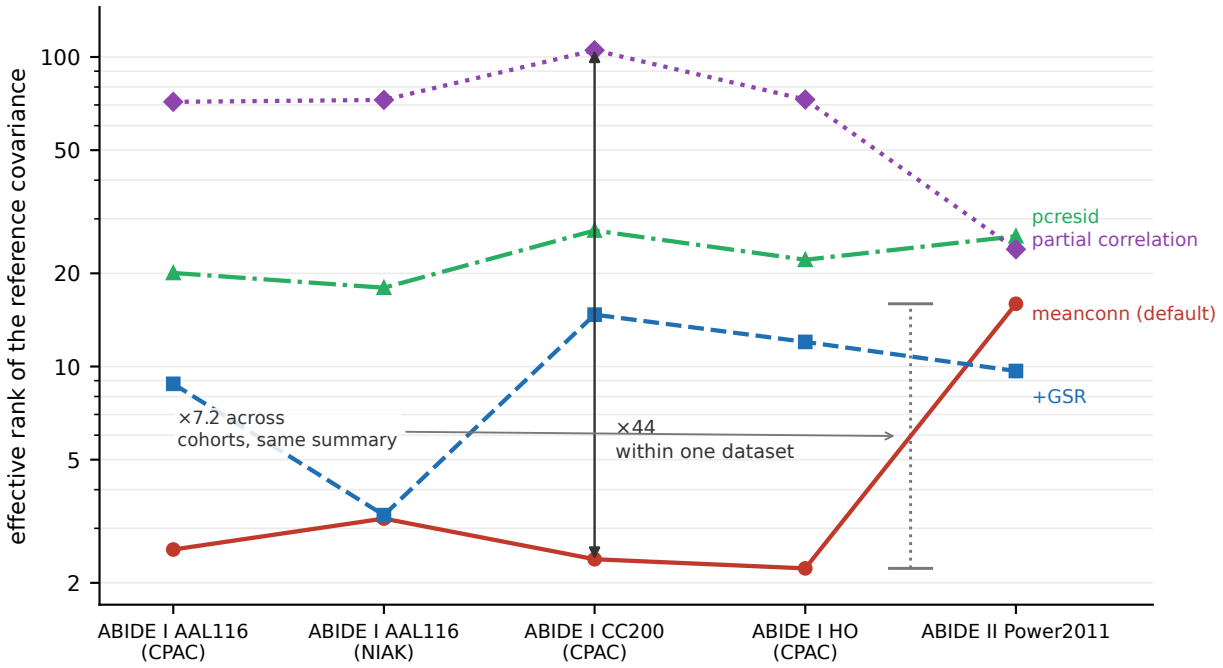


Figure 2: The ROI summary, not the cohort, sets the geometry of the feature space. Effective rank $(\sum \lambda_i)^2 / \sum \lambda_i^2$ of the reference cohort's correlation matrix, one line per ROI summary definition, across five dataset/parcellation combinations (logarithmic axis; exact values in Table 1). Changing the summary moves the effective rank by up to a factor of 44 within a single dataset; changing the dataset or parcellation while holding the summary fixed moves it by at most a factor of 7.2. The convention is the binding choice.

we add is the per-subject, per-feature quantification—effective rank as a single number, and the distribution of R^2 over features—and the consequence in a variable-selection setting.

The choice is not a refinement. The four summaries are almost mutually uncorrelated at the ROI level (Table 2): median per-ROI Pearson correlation between 0.00 and 0.26, with *every one* of 113 ROIs below 0.5. For comparison, the two preprocessing pipelines compared in §3.10—a coarser manipulation, applied to the same subjects in the same atlas—agree at median $r = 0.436$. A reader choosing a summary is choosing which measurement to analyse, and that choice displaces the features further than any subsequent statistical decision we examine in this paper. The constructive version of this observation is known in the prediction literature: no single functional-connectivity measure consistently outperforms the others, and concatenating several is more consistent than any one (Mohanty et al., 2020).

The conventional neuroimaging remedy is not reliable. Global-signal regression, the field's standard response to the global component, raises the effective rank on ABIDE I CPAC (2.56 → 8.79), does nothing on the NIAK pipeline (3.23 → 3.31), and *lowers* it on ABIDE II (15.95 → 9.67). The contrast with the ABIDE I direction is robust: over five random reference-cohort splits the ABIDE I effect is $2.49 \pm 0.08 \rightarrow 9.08 \pm 0.47$ and the ABIDE II effect is $17.20 \pm 1.18 \rightarrow 11.18 \pm 0.52$. It is not a rank-ceiling artefact—ABIDE II has $T \geq 119$ for every subject against $p = 94$, so no subject is in the $T \leq p$ regime—and we do not have a mechanism for it.

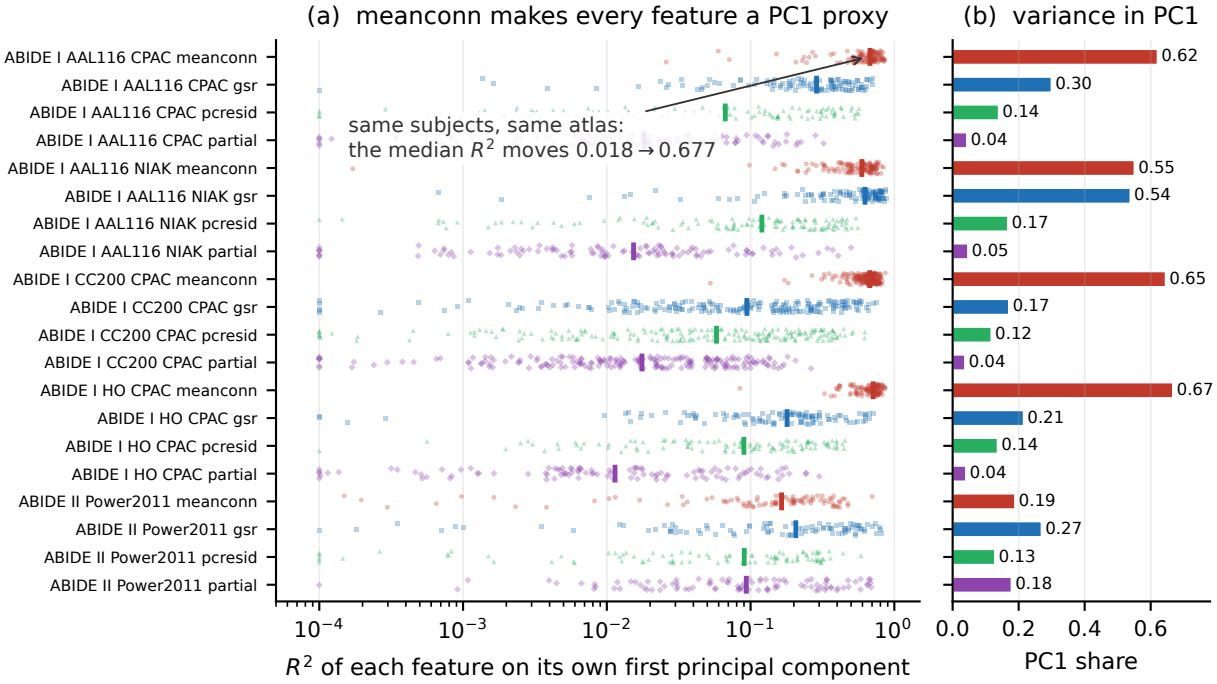


Figure 3: The conventional summary makes every feature a proxy for the leading principal component; partial correlation does not. (a) One row per dataset/summary configuration (logarithmic axis); each point is one ROI's R^2 on its own principal component, computed on the reference cohort, and the vertical bar marks the median. Holding subjects and atlas fixed (ABIDE I, AAL116, CPAC) the median moves from 0.018 for partial correlation to 0.677 for meanconn. ABIDE II is shown as it is: its conventional summary sits at a median of 0.164, which is the cohort-dependence that §3.11 takes up. (b) The same configurations by PC1 share.

pair of summaries (ABIDE I, AAL116, CPAC)	median per-ROI r	min	ROIs with $r < 0.5$
meanconn vs. GSR	0.219	−0.109	113/113
meanconn vs. pcresid	−0.129	−0.317	113/113
meanconn vs. partial	0.210	0.014	113/113
GSR vs. partial	0.005	−0.205	113/113
partial vs. pcresid	0.114	−0.026	113/113

Table 2: Agreement between ROI summary definitions, per ROI, over the same subjects. The summaries are effectively different measurements.

We flag it because it runs opposite to the published expectation that removing the global signal removes the dominant mode (Moujaes et al., 2024), and because the direction of a GSR effect on connectivity structure is contested on other grounds: GSR is known to create group differences that favour one group, remove the strongest true differences, and displace the location of group effects (Gotts et al., 2013; Saad et al., 2012; Ciric et al., 2017), while its motion-attenuation benefit is real (Ciric et al., 2017; Murphy and Fox, 2017). The defensible reading is that GSR is a choice with consequences in both directions rather than a correction. Removing the leading principal components worked in every configuration we tried.

3.2 The received explanation fails its own test

If the spectral collapse caused the null results, a simulation using the collapsed covariance should return nothing. It does not. Table 3 sweeps the covariance from the ABIDE structure to the identity, and across AR(1) structures and the twenty real feature matrices of Table 1, holding the pipeline fixed. Figure 4a,b shows the same sweep on the effective-rank axis, where the invariance is visible at a glance.

		effective rank	power	FDP
A: $\Sigma(t) = (1-t)\Sigma_{\text{ABIDE}} + tI$	$t = 0$	2.56	**0.859**	0.154
	$t = 0.10$	3.15	0.974	0.132
	$t = 1$	113.0	1.000	0.176
B: AR(1), $\rho \in [0.02, 0.80]$	—	25–113	1.000	0.17–0.20
C: 20 real feature matrices	—	2.23–105	0.770–1.000	0.15–0.21

Table 3: Power and realised FDP against covariance conditioning, pipeline held fixed ($n_{\text{ref}} = 393$, $n_{\text{main}} = 459$, $q = 0.20$, $c = 0.05$, $k = 12$, 200 replications; every configuration fully converged). Using the real ABIDE covariance as ground truth gives power 0.859, not zero. Across all 41 configurations the minimum power is 0.770, attained by the NIAK + GSR combination that Table 1 identifies as the one case where GSR fails to help.

The spectral collapse is real and it costs something—0.859 against 1.000, about fourteen percentage points—but it does not make the procedure powerless. Whatever produced the null results on real data, it was not the conditioning of the covariance. This matters because the two readings are easy to hold at once and they contradict each other: the same covariance can be described as degenerate enough that no construction should work, and yet deliver per-signal power of 0.859; those two statements cannot both hold, and this experiment is the resolution.

3.3 What limits power is the number of true signals

The same conclusion holds when signals are injected into the *real* covariates. Table 4 reports per-signal power by covariance arm and support size; Figure 4c shows the same contrast against the strength of the injected signal, which is the factor that does move power.

support	conditional $\lvert t\rvert$	ref ($n = 100$)	ref ($n = 393$)	main as its own ref
random, $k = 12$	3	0.59	0.68	0.69
	4	0.79	0.88	0.89
	6	0.93	**0.98**	**0.98**
random, $k = 3$	6	0.16	0.21	0.19
near-collinear, $k = 3$	6	0.13	0.28	0.24

Table 4: Per-signal power (fraction of injected coordinates recovered) on real ABIDE I covariates, by covariance arm. Enlarging the reference cohort has a small monotone effect that is visible at moderate signal—at conditional $|t| = 3$, $k = 12$, per-signal power rises from 0.590 at $n_{\text{ref}} = 100$ to 0.676 with the full reference cohort—and narrows as the signal strengthens, which is the direction the theory predicts since a noisier $\hat{\Sigma}$ is the more conservative one (Candès et al., 2018). So **the reference cohort is a secondary constraint, not the binding one**. At fixed covariance and signal strength, the dominant factor is k. The last column is a *reference point, not a recommended procedure*: it estimates $\hat{\Sigma}$ from the same covariates that enter the statistic, so it does not meet the known-F_X requirement and its FDR control is not guaranteed by the theory. Its realised FDP (0.04–0.15, inside $q = 0.20$) is nonetheless unsurprising, because Barber et al. (2020) bound the inflation by the divergence between the true and working conditional distributions, and that divergence here has one source—finite-sample estimation—rather than the two the reference-cohort arms carry. We use it to ask what reference data buys in power, and nothing else. The near-collinear support—the three ROIs with the highest variance-inflation factors in the atlas—does no worse than a random support of the same size, so the degeneracy of the features does not limit power in the specific directions we can measure.

Two facts emerge. Reference-cohort size has only a small effect, and only where the signal is weak enough for $\hat{\Sigma}$'s noise to matter: at conditional $|t| = 3$, $k = 12$, per-signal power is 0.590 at $n_{\text{ref}} = 100$ against 0.676 for the full cohort, and the three arms converge to 0.93/0.98/0.98 by $|t| = 6$. The factor that does move power is the number of true signals, from 0.97 at $k = 12$ to 0.20 at $k = 3$ under identical signal strength. This is knockoff+'s own conservatism: the threshold in (5) requires $R \geq (1 + F)/q$, which is hard to satisfy when few coordinates are non-null. It is a property of the method, not of the covariance and not of the reference cohort. The dependence is qualitatively consistent with the first-passage characterisation of the knockoff+ threshold (Zhang, 2026); we note that the asymptotic power theory in Weinstein et al. (2022) assumes the number of non-nulls grows with p, a regime our $k = 3$ configurations are outside.

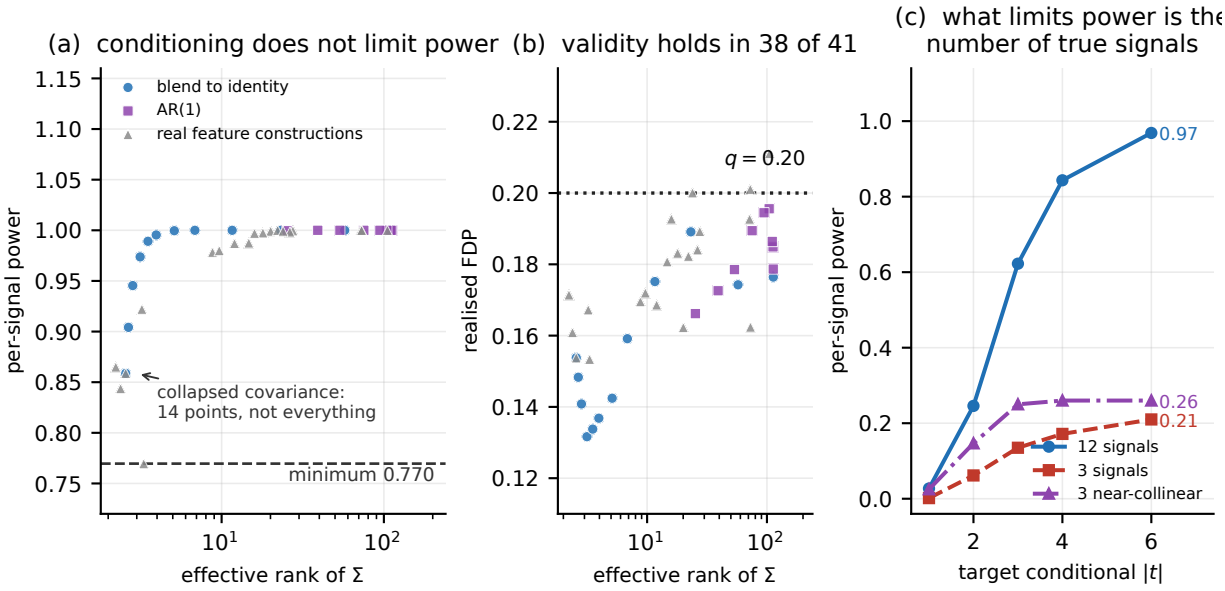


Figure 4: Conditioning does not limit power; the number of true signals does. (a) Per-signal power against the effective rank of Σ over all 41 covariance configurations of Table 3 and three construction families, spanning effective rank 2.2 to 113; the dashed line is the minimum over the whole set. The most collapsed covariance costs 14 points of power—it does not return nothing. (b) The realised FDP over the same configurations; the guarantee holds in 38 of 41, so validity is not the constraint here either. (c) The factor that does move power: per-signal power against the target conditional $|t|$ of the injected signal, for 12 random signals, 3 random signals, and 3 near-collinear signals. The number of true signals, not the conditioning, sets the ceiling. Panels are on different x-scales; (a) and (b) share one.

3.4 Exhausting the alternatives

If the limitation is not the reference cohort, it is worth establishing that it is not any of the other obvious candidates either. We tested three classes of alternative. The first two—knockoff constructions and Gaussian mirrors—share one positive control and are compared in Figure 5 on the two axes that matter at once, power and validity; the third is treated separately below because it reports selections rather than an error rate.

Knockoff constructions. Table 5 reports the construction diagonal s for six methods on degenerate and non-degenerate features. On the degenerate features the six constructions are indistinguishable: median s between 0.063 and 0.088, implying $\mathrm{Corr}(X_j, \tilde{\mathbf{X}}_j)$ between 0.912 and 0.937. Every construction is forced into the same near-replica regime, and the SDP solution is not better—it minimises s on some coordinates, returning $s = 0$ for the coordinates it sacrifices. This is the rank deficiency that Spector and Janson (2022) identify, appearing here as a ceiling on what any construction can achieve when Σ is nearly singular. On the less-degenerate features the construction matters a great deal: on ABIDE II partial-correlation features, SDP raises s to 4.6 times the equicorrelated value.

	median s (and $\mathrm{Corr}(X_j, \tilde{\mathbf{X}}_j) = 1 - s$)					
configuration	equi	identity	SDP	ASDP	MVR	ME
AAL116 meanconn ($\lambda_{\min} = 0.032$)	0.064	0.064	0.088	0.063	0.069	0.081
ABIDE II partial ($\lambda_{\min} = 0.067$)	0.133	0.133	**0.611**	0.294	0.431	0.511

Table 5: Knockoff constructions compared at matched data. On degenerate features all six are indistinguishable, and SDP returns $s = 0$ on the coordinates it sacrifices. On less-degenerate features the choice matters by a factor of 4.6.

A method that never inverts Σ. Gaussian mirrors (Xing et al., 2023) construct, for each variable, a pair of mirror variables $x_j^{\pm} = x_j \pm c_j z_j$ with $z_j \sim N(0, I_n)$ and c_j chosen so that the mirror statistic is exactly symmetric under the null. The construction never forms Σ^{-1}, so the spectral collapse cannot enter it. On our features it is *valid*—under a global null it returns a nonempty selection in 8.0% of replications, within $q = 0.20$—but it recovers only 0.421 of injected signals against 0.977 for the knockoff filter, at FDP 0.000 against 0.110. The reason is measurable and is the same geometry: c_j is set by the residual of x_j on all other columns, whose norm is 6.60 against $\|x_j\| = 21.4$, so each variable is 90.5% explained by the others. A method designed to avoid Σ^{-1} fails at the same place, which is direct evidence that the inversion is not the binding constraint.

Cross-environment consistency. The multi-environment knockoff filter (Li et al., 2022) tests not "is this ROI associated with diagnosis" but "in how many environments is it consistently associated", with FDR control. This matches our setting, where the strongest observation is that nothing is consistent. Using the eight ABIDE I sites with $n \geq 40$ and both diagnostic classes as environments, and estimating Σ from the remaining environments in each case, no ROI is selected at any consistency level $r \in \{1, \ldots, 5\}$, and a negative control that permutes the response within each environment also returns nothing.

That zero is uninformative, and we established why before reading it as a result. Unlike every other alternative in this section, this arm has a positive control: we injected a signal that is consistent across environments by construction—the same twelve coordinates, the same sign and

the same standardised effect size in every site, calibrated to a conditional $|t|$ of 6 as elsewhere in this paper—and ran the filter on it over 60 replicates. In the configuration above the filter recovers a mean of 0.100 of the injected coordinates at best ($r = 2$, $c = 0.05$) and nothing at all at $c = 0.15$, while the permuted control returns nothing in either case. The limitation is the size of the environments rather than the filter: pooling the same subjects into four groups of about 213 rather than eight sites of 42–169, the identical injected signal is recovered at a mean rate of 0.762 at $r = 1$ and 0.736 at $r = 2$. With $p = 113$ and every site holding fewer than 170 subjects, the per-environment knockoff fits are too weak to yield a signed statistic. We therefore read the empty selection on the real response as showing only that this arm cannot distinguish "no replicable association" from "no power"; no other arm in this paper has that property, and the two failures of §3.6 and §3.7 are established from controls that do have power.

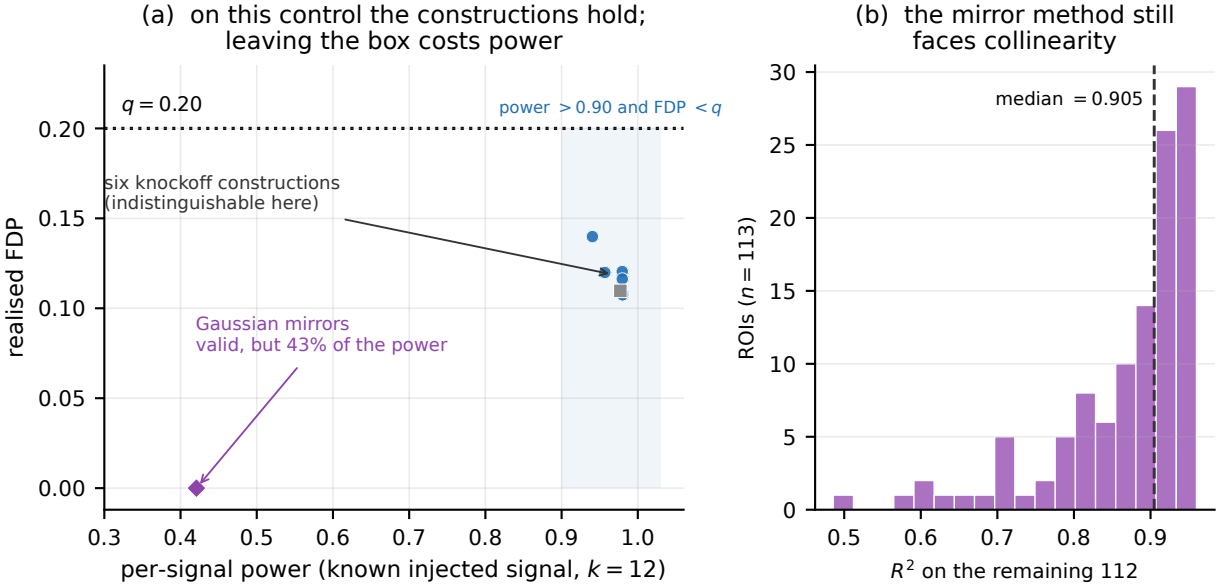


Figure 5: Alternative constructions do not escape the trade-off. (a) Per-signal power against realised FDP for the six knockoff constructions and the Gaussian mirrors, all evaluated on one positive control (an injected signal of known conditional strength, $k = 12$, calibrated on the same quantity for every arm); the shaded box is the region with power above 0.90 and FDP below q. The six constructions are indistinguishable inside it. Gaussian mirrors leave it: they are valid—they never invert Σ—but recover 43% of the power. The matrix-power family is not on this panel because it uses a different injection convention (§3.5); there the constructions do leave the box, and what they lose is validity rather than power. (b) The mechanism behind the mirrors' shortfall: R^2 of each ROI on the remaining 112, median 0.905, so avoiding Σ^{-1} does not avoid collinearity.

3.5 The matrix-power family: reported error and true error move in opposite directions

The three alternatives above are discrete. A continuous family makes the same point more sharply. If Σ is a correlation matrix, the linear transform

$$\mathbf{X}_t \;=\; \mathbf{X}\,\Sigma^{(t-1)/2} \qquad \Longrightarrow \qquad \operatorname{corr}(\mathbf{X}_t) = \Sigma^t, \tag{7}$$

since $\Sigma^{(t-1)/2}\,\Sigma\,\Sigma^{(t-1)/2} = \Sigma^t$. Applied identically to the reference and main cohorts, this is a legitimate one-parameter family of feature representations, interpolating between the marginal correlation ($t = 1$), the whitened features ($t = 0$), and the precision ($t = -1$). The effective rank moves monotonically across it, from 2.56 to 111 and back to 76.

This family is a natural way to try to increase the discriminability of degenerate features, and it is worth reporting what happens when one does. Figure 6 and Table 6 give the result under three responses: an injected signal fixed in the *original* ROI space, the real diagnosis, and a global null.

	injection, $k = 12$		real response		global null	
t	eff. rank	power / FDP	$P(\hat{S} \neq \emptyset)$	reported	$P(\hat{S} \neq \emptyset)$	true FDP
1.0	2.56	0.892 / 0.130	0.355	0.250	0.055	1.00
0.6	17.63	0.957 / 0.493	0.540	0.200	0.045	1.00
0.2	97.93	0.965 / 0.808	0.790	0.143	0.075	1.00
0.0	111.09	0.954 / 0.826	0.875	0.125	0.100	1.00
-0.6	94.70	0.881 / 0.766	0.980	0.091	**0.280**	1.00
-1.0	76.10	0.790 / 0.726	0.985	0.100	0.245	1.00

"reported" is the median of the procedure's own threshold ratio $\min_t(1 + F_t)/R_t$; under a global null the true FDP is 1 whenever anything is reported, so $P(\hat{S} \neq \emptyset)$ *is* the expected FDP and must be compared against $q = 0.20$.

Table 6: The matrix-power family (7). Power on a fixed injected signal is flat across the whole range (0.79–0.97); what moves is the false discovery count. The two monotone columns tell the story: the procedure's own reported error improves from 0.250 to 0.091 while the true error rises from 0.130 to 0.826, and under a global null the rate of nonempty selections crosses the nominal level between $t = -0.2$ and $t = -0.4$.

Three readings, in increasing order of importance. The transform does not help: power is flat, so it does not make true signal easier to find. It hurts in a way that is invisible: at $t = 0$ the procedure reports a threshold ratio of 0.125, comfortably inside q, while committing a false discovery proportion of 0.826. And the global-null column closes the argument without appealing to any biological ground truth: with a response drawn independently of the covariates, the rate of nonempty selections rises from 0.055 at $t = 1$ to 0.280 at $t = -0.6$, crossing $q = 0.20$; since every discovery in that column is false by construction, this is a direct violation of the guarantee. A user tuning t by the two quantities the software shows them—number of discoveries and reported ratio—would be led to $t \approx -0.6$, where the true error is 0.77.

3.6 Cohort shift breaks validity without reducing power

The simulations above draw both cohorts from one Σ, which measures estimation error only. The subject of this paper is a reference cohort that is *different*. We draw $\mathbf{X}_{\text{ref}} \sim N(0, \Sigma)$ and $\mathbf{X}_{\text{main}} \sim N(0, \Sigma_\delta)$ with $\Sigma_\delta = \text{pd}\big((1-\delta)\Sigma + \delta P\big)$ for a sparse random perturbation P.
The failure mode is specific (Figure 7). As the cohorts diverge, power *rises* from 0.859 to 0.999 while the realised FDP rises from 0.154 to 0.422. Apparent power grows because knockoffs built

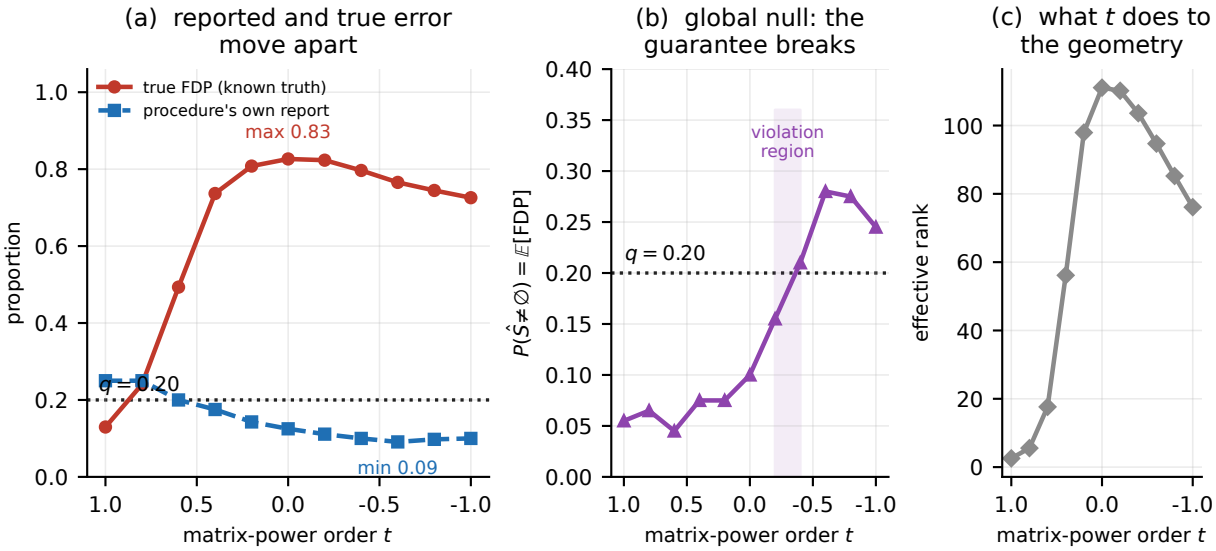


Figure 6: Along the matrix-power family the procedure's own report and its true error move in opposite directions. (a) Under an injected signal whose truth is known, the realised FDP (red, solid) rises to 0.826 while the procedure's own reported threshold ratio (blue, dashed) falls to 0.091; both are proportions, so they share an axis. (b) Under a global null the true FDP is 1 whenever anything is reported, so the curve *is* the expected FDP; it crosses $q = 0.20$ between $t = -0.2$ and $t = -0.4$ (shaded). (c) What t does to the geometry, so that the family can be read on the effective-rank scale of §3.1. Series are distinguished by marker and line style as well as colour, so the figure survives grayscale printing.

δ	mismatch	power	realised FDP
0.00	0.000	0.859	0.154
0.10	0.062	0.966	0.144
0.20	0.124	0.994	0.180
0.30	0.186	1.000	**0.227**
0.80	0.497	0.999	**0.422**

Table 7: Cohort distribution shift. Mismatch is the RMS difference of the off-diagonal correlation entries. The observed ABIDE I mismatch between the main and reference cohorts is 0.0593, corresponding to $\delta \approx 0.10$ (row 2), where FDP is 0.144.

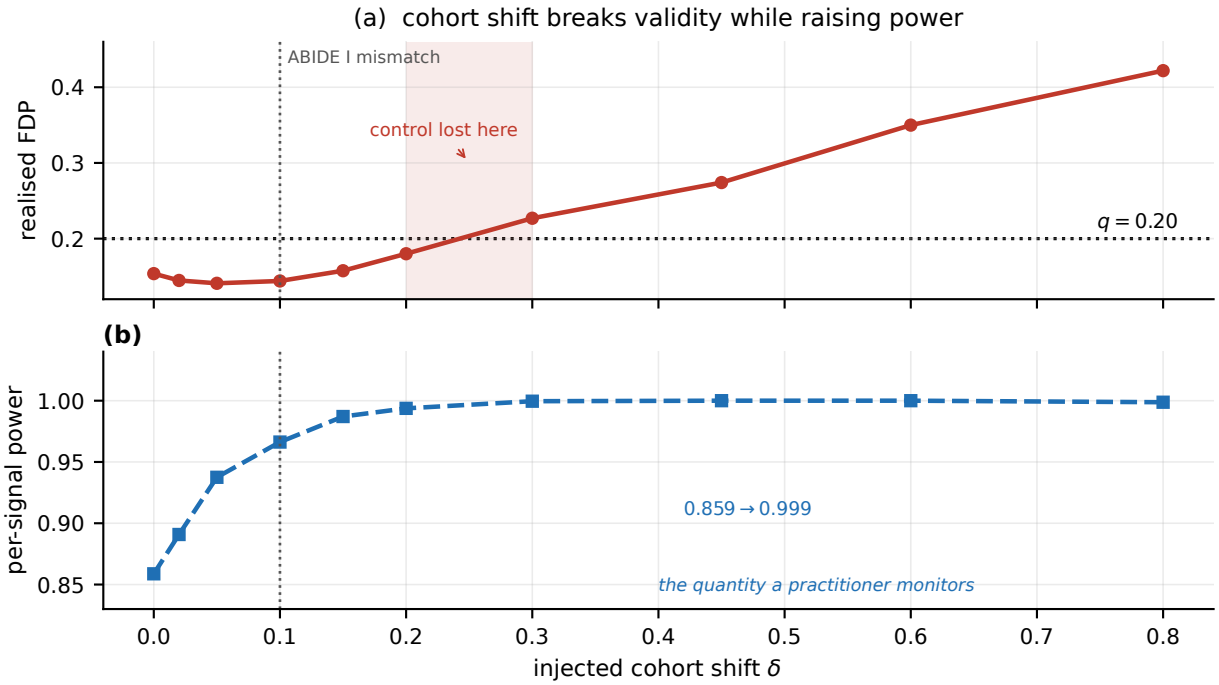


Figure 7: Cohort shift breaks validity while raising power. (a) Realised FDP against the injected cohort shift δ; the shaded band marks the interval between the last level at which FDP is controlled and the first at which it is not, and the dotted line marks the grid level whose simulated mismatch, 0.062, is closest to the 0.059 observed in ABIDE I. (b) Per-signal power over the same shift, which rises throughout. A user monitoring the quantity in (b) reads this failure as improvement; the failure is silent because the quantity they watch moves the wrong way.

from a mismatched covariance are less correlated with the originals, so the statistic separates them more easily; the same mechanism destroys the calibration. A user watching only the discovery count reads this as improvement. FDP control is lost between $\delta = 0.20$ and $\delta = 0.30$ (mismatch 0.124–0.186; we did not resolve the crossing more finely than the grid), and the observed ABIDE I mismatch of 0.0593 sits inside the safe region—which is why this failure mode does not explain our null results either. It would have produced *more* discoveries, not fewer.

The precise condition this violates is worth naming. Knockoff validity requires exchangeability under the conditional law $L(\mathbf{X} \mid Y)$, whereas a reference cohort supplies a distribution valid under the marginal $P(\mathbf{X})$; when the sampling law differs from the population law the implication breaks (Zhao et al., 2025). This is the imaging analogue of the reference-panel mismatch that GWAS knockoff methods must contend with (Chen et al., 2021; Kanai et al., 2022; Benner et al., 2017).

3.7 The penalty constant must satisfy two conditions

The fixed-λ statistic (6) has one tuning parameter, and we chose it badly at the outset of this analysis. Holding data, knockoffs, covariance and solver fixed and varying only c (Figure 8):

	constraint 1: the signal must survive		constraint 2: the solver must converge
c	recovered (of 12)	all 12 recovered	non-convergence, real response
0.02	11.97	0.97	96.7%–100%
0.03	11.97	0.97	**13.3**%–**46.7**%
0.05	11.73	0.80	**0**%
0.10	6.95	0.13	0%
0.15	**1.12**	0.02	0%
0.30	0.07	0.00	0%

Table 8: The penalty constant must satisfy two constraints at once, and they pull in opposite directions. Constraint 1 is measured by injecting signals of known conditional strength ($k = 12$, conditional $|t| = 6$, 60 replications) and counting how many coordinates are recovered; constraint 2 is measured on the real (weak) response, where a small penalty yields a dense lasso and coordinate descent may exhaust its iteration budget. Neither constraint is reported by the software. Only $c = 0.05$ satisfies both.

A signal whose conditional $|t|$ reaches 6 is almost entirely penalised away at $c = 0.15$: 1.12 of 12 coordinates are recovered on average, and all twelve are recovered in only 2% of replications. Reducing c fixes that, but on the real response the penalty also controls how dense the solution is, and the real response is weak: at $c = 0.02$ the solver exhausts the iteration cap in 96.7%–100% of runs and at $c = 0.03$ in up to 46.7%, returning a statistic that has not converged and should not be interpreted. Only $c = 0.05$ both recovers the injected signal (11.73 of 12) and converges on the real response, with 2.4× iteration headroom.

We note that this is a regime our power theory does not cover. Asymptotic analyses of the penalty-power relationship assume an i.i.d. design and often recommend cross-validated tuning

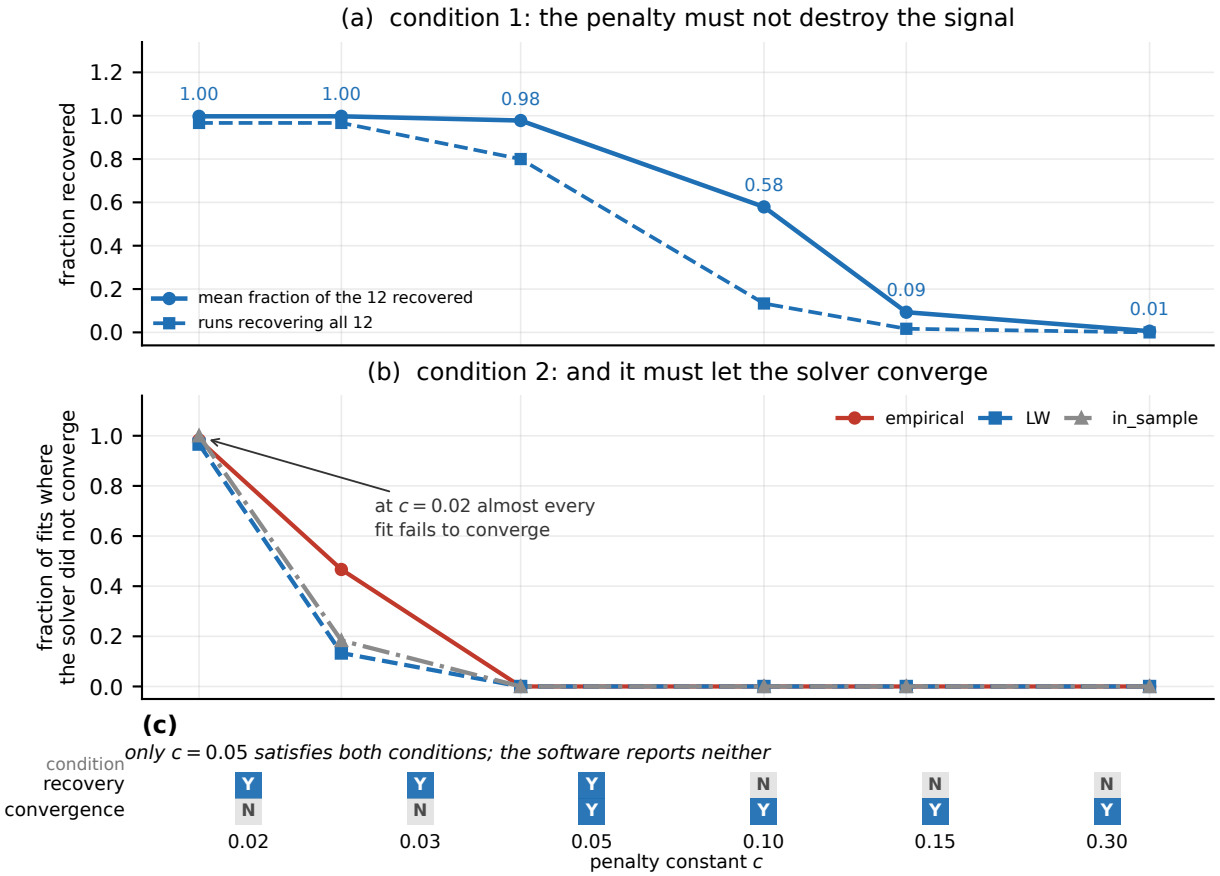


Figure 8: The penalty constant must satisfy two conditions at once, and they pull in opposite directions. (a) Fraction of the 12 injected signals recovered, against c (logarithmic axis), shown both as the mean fraction recovered (solid) and as the fraction of runs in which all twelve are recovered (dashed); both collapse beyond $c = 0.05$. (b) Fraction of fits in which the solver failed to converge on the real response, for the empirical covariance and for two Ledoit–Wolf variants (§2.5); below $c = 0.05$ almost every fit fails. (c) The conjunction, taking recovery as a mean fraction of at least 0.90: only $c = 0.05$ satisfies both conditions, and the software reports neither, so the failure is invisible unless both are checked deliberately.

(Weinstein et al., 2022); on data with $p/n = 0.25$ and a highly correlated Σ the relation is not monotone in the way those results assume, and the relevant fragility has been documented for the one-standard-error rule (Candès et al., 2018).

3.8 Real-data answers are distributions, not numbers

Knockoff generation is random. It is tempting, and common, to report the discovery count from a single draw; the numbers below show why that is an artefact. A single draw returning $\hat{S} = \emptyset$ reads as "yields no discoveries", while the same configuration measured over repeated draws returns nothing in 64.5% of them and up to 28 regions in the rest. The two statements cannot both stand, and the first is the one an author is more likely to report.

Holding everything except the draw fixed ($c = 0.05$, 200 draws per configuration), the real-response results are (Figure 9):

dataset	summary	$P(\hat{S} \neq \emptyset)$	median $\lvert\hat{S}\rvert$	max $\lvert\hat{S}\rvert$	median ratio	converged
ABIDE II, Power2011	**pcresid**	**0.815**	**12**	40	**0.148**	1.00
ABIDE I, Harvard–Oxford	meanconn	0.420	0	24	0.250	1.00
ABIDE I, AAL116, CPAC	meanconn	0.355	0	28	0.250	1.00
ABIDE II, Power2011	meanconn	0.190	0	20	0.369	1.00
ABIDE I, AAL116, CPAC	partial	0.010	0	5	0.742	1.00
ABIDE I, AAL116, NIAK	GSR	0.000	0	0	0.857	1.00

Table 9: Distribution of knockoff selections on the real ASD/control response, 200 independent draws per configuration with the pipeline otherwise fixed. The reported ratio is $\min_t(1 + F_t)/R_t$, the optimal attainable threshold ratio. One of the twenty configurations we ran (CC200 with meanconn features, convergence rate 0.37) is excluded for non-convergence; all rows shown are fully converged. **These are not discoveries**: on real data we have no ground truth for which ROIs are genuinely associated with diagnosis, so the selections cannot be validated and we make no claim about them. They are reported because the *distribution* is the honest output of the procedure, and because the spread across configurations is itself the finding.

Two readings. First, "no discoveries" is not a property of this data: the conventional meanconn features on AAL116 return a nonempty selection in 35.5% of draws and as many as 28 regions, and the Harvard–Oxford parcellation is higher still. A single-draw analysis would have justified the opposite conclusion, which is why we report the whole distribution. Second, the output depends on the feature definition far more than on anything else we varied: from $P(\hat{S} \neq \emptyset) = 0.000$ for NIAK GSR to 0.815 for ABIDE II pcresid. The procedure's own reported ratio ranges from 0.148 (inside q) to 0.857 (far outside), so the *feasibility* of a controlled selection at $q = 0.20$ is also feature-dependent. We emphasise again that the 0.815 configuration is not a discovery claim: it is a claim about what the procedure reports, from a procedure whose report we have shown to be unreliable under other manipulations (§3.5, §3.6).

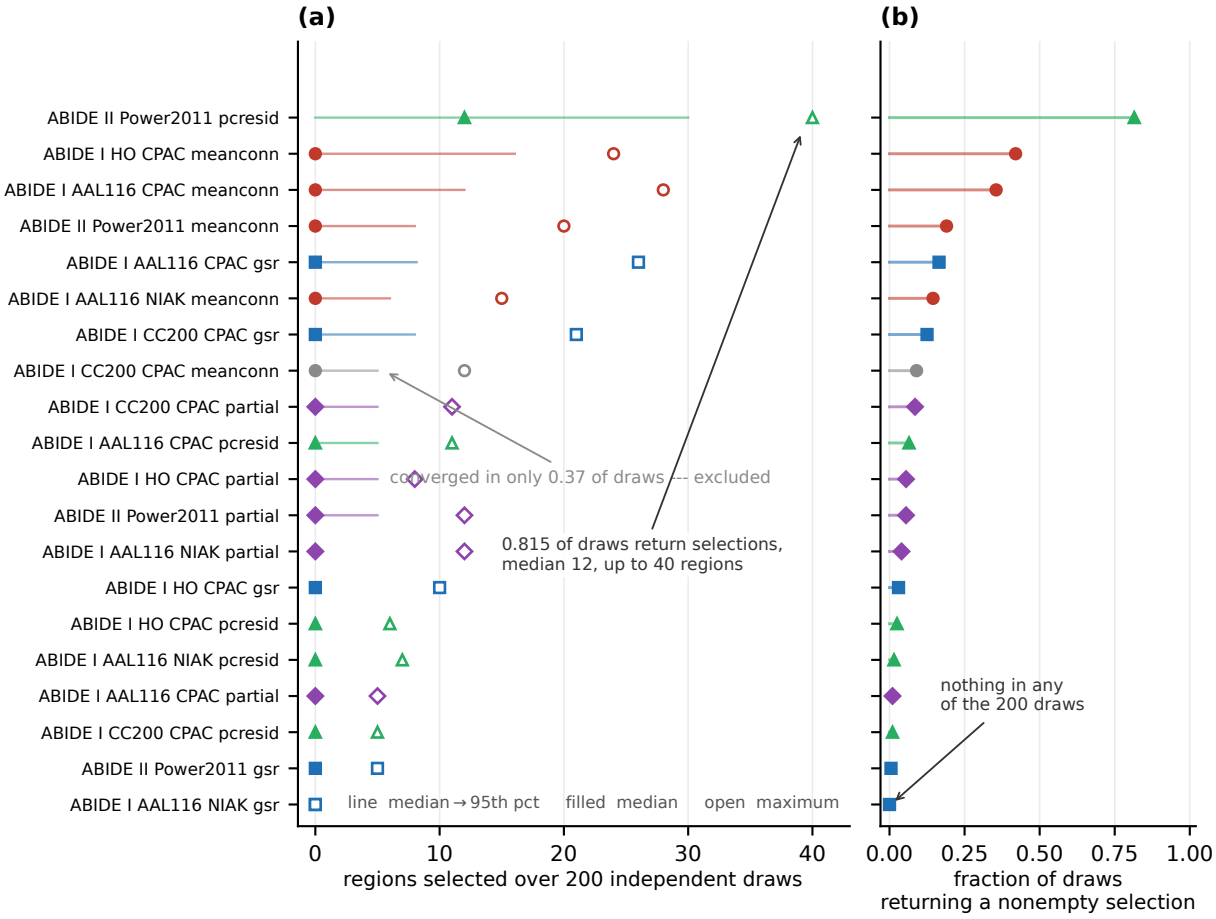


Figure 9: The output of the procedure on the real response is a distribution, not a number. (a) One row per dataset/summary configuration, ordered by the fraction of draws returning a nonempty selection; the thin line spans the median to the 95th percentile of $|\hat{S}|$, the filled dot marks the median and the open dot the maximum over 200 independent knockoff draws. The grey row is excluded: its solver converged in only 0.37 of draws. Of the twenty configurations, nineteen have a median of zero; eighteen of them nevertheless reach a nonzero maximum, and one returns nothing in any draw. (b) The same rows, showing that fraction directly. The top row returns selections in 0.82 of draws and up to 40 regions; the bottom row returns nothing in any draw—same procedure, same response, same q, a different summary. $P(\hat{S} \neq \emptyset)$ equals the fraction of draws whose optimal ratio falls below q, so panels (a) and (b) encode one quantity, not two. These are not discoveries: on real data no ground truth exists, so the spread across configurations is itself the finding.

3.9 Two properties supported by controls

Two further properties follow from the same controls and are worth stating separately, because both are easy to get wrong.

Aggregating draws by averaging the statistics is invalid. Median aggregation across M independent knockoff draws removes the zero-discovery runs entirely and more than doubles mean pairwise Jaccard similarity, so it appears to work. Under a global null it returns a median of 20.6 discoveries at $M = 10$ with a realised FDP of 0.97, while the e-value construction of Ren and Barber (2024),

$$e_j = p \cdot \frac{\mathbf{1}\{W_j \geq T\}}{1 + \#\{k : W_k \leq -T\}}, \tag{8}$$

with e-BH (Wang and Ramdas, 2022) applied to the averaged e-values, returns nothing—the correct answer. On a single draw, e-BH applied to (8) reproduces the knockoff+ selection set identically in 25 of 25 instances, an algebraic identity and therefore a unit test of the implementation rather than evidence of validity.

The reason is worth recording, because the argument for median aggregation is superficially convincing. For a null j each draw's $W_j^{(m)}$ is symmetric about zero with an independent fair sign, so the *marginal* distribution of the median is symmetric. Knockoff+ requires the null signs to be jointly independent given the magnitudes, and the median does not inherit that: its sign and magnitude are entangled through the pattern of signs across draws. Marginal symmetry is necessary and not sufficient, and a numerical check of marginal symmetry cannot detect the difference. This failure is corroborated in the aggregation literature (Nguyen et al., 2020). We note a distinction that matters for interpretation: the median we criticise is taken *across independent knockoff draws*. Medians taken *within* a single draw across simultaneous knockoffs are a different object and not subject to this argument.

The discovery set depends on the statistic's configuration. Holding data, covariance and knockoff draw fixed, three-fold cross-validation selects a substantially weaker penalty than five or ten ($C = 0.616$ against 0.234) and returns seven regions where they return none; all three runs converge and none emits a warning. The iteration cap behaves similarly. Unlike the draw-to-draw instability, this is a function of choices the analyst makes, none of which the procedure reports, and aggregating over draws does not help when the disagreement is between fold counts.

3.10 Agreement across preprocessing pipelines

ABIDE I distributes each subject through two independent pipelines (`cpac` and `niak`), both emitting AAL116 parcel time series in the same ROI order, so we can vary the pipeline alone with every subject appearing on both sides.

level of the analysis	cross-pipeline agreement	
per-ROI feature vector	median $r = 0.436$	(90/113 below 0.5)
between-group t-statistic	$r = 0.590$	
knockoff W ranking	$r = 0.318$ (empirical), 0.237 (LW)	

The two pipelines retain different numbers of volumes from the same acquisition (196 against 164 for a representative subject; 146 against 55 in the worst case), and their parcel time series are on entirely different scales (mean 0.12, s.d. 13.8, against mean 406, s.d. 164). Correlation is scale-invariant so the comparison is legitimate, but a reader assuming that "the same subject in the same atlas" implies comparable inputs would be wrong by a wide margin.

The spectral mechanism survives the pipeline change: under both pipelines the empirical reference covariance is the more nearly singular and shrinkage raises it (CPAC $\lambda_{\min}$ 0.0218 $\rightarrow$ 0.0324; NIAK 0.0172 $\rightarrow$ 0.0397). Nothing downstream does. This is the same finding as §3.1 one level upstream: choices that precede the statistics displace the features by as much as anything downstream.

3.11 A third cohort: the ordering replicates, the constant does not

Every result so far comes from autism cohorts. We therefore added **ADHD-200** (Athena preprocessed release, AAL116, $p = 116$, $n = 664$; 427 typically developing against 237 ADHD across nine sites), a different disorder with different participants, scanners and acquisition protocols (Table 10).

	effective rank of meanconn			cohort mismatch	
cohort	n_{ref}=80	121	300	observed	noise floor
ABIDE I, AAL116	2.96	2.95	2.60	0.059	0.048
ABIDE II, Power2011	12.89	13.88	15.56	0.105	0.093
ADHD-200, AAL116	6.55	**6.65**	—	**0.278**	**0.160**

Table 10: Third-cohort validation. The conventional summary is the most degenerate in all three cohorts, but its severity is cohort-specific: at matched reference size the effective rank spans 2.95 to 13.88. The noise floor is the mismatch obtained by randomly splitting each cohort into two groups of the same sizes, which by construction differ only by sampling; the excess of the observed value over it is the genuine distributional difference.

The ordering replicates; the severity does not. In ADHD-200, as in both ABIDE cohorts, the conventional meanconn summary has the lowest effective rank and partial correlation the highest (6.65 against 25.98 for pcresid and 40.20 for partial). But the *severity* is cohort-specific: at matched reference-cohort size ($n_{\text{ref}} = 121$) the effective rank of the same summary is 2.95 in ABIDE I and 6.65 in ADHD-200, and 13.88 in ABIDE II. That is exactly what §3.1 predicted—the diagnostic is portable, the constant is not—and it is the reason we recommend reporting the diagnostic rather than a threshold.

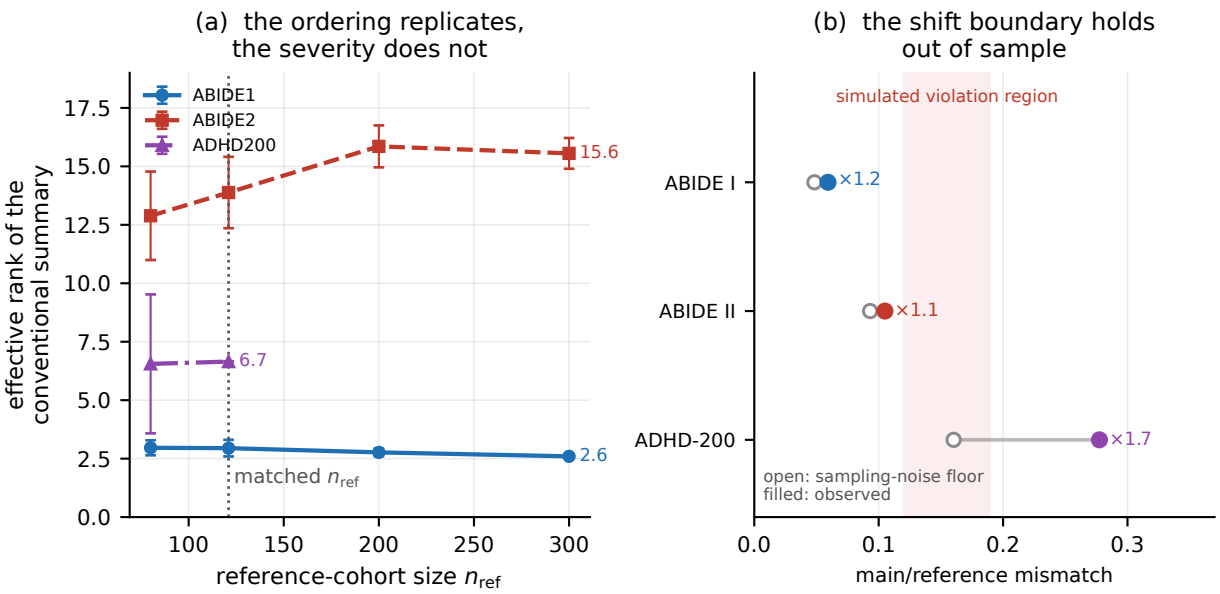


Figure 10: Third-cohort validation: the ordering replicates, the severity does not. (a) Effective rank of the conventional summary against matching reference-cohort size (error bars, one standard deviation over draws); at $n_{\text{ref}} = 121$ the same summary spans 2.95 (ABIDE I) to 13.88 (ABIDE II), so the ordering of summary definitions replicates while their severity does not. (b) Measured main/reference mismatch (filled dot) against the sampling-noise floor obtained by randomly splitting each cohort into two groups of the same sizes (open dot); the leader line gives their ratio. ABIDE I and II sit at 1.2× and 1.1× their floors, so their mismatch is almost entirely sampling; ADHD-200 sits at 1.7×. The panel plots the observed mismatch against the floor, not their difference: the excess 0.117, which is what lands at the lower edge of the simulated violation region (shaded), is obtained only after the floor is subtracted. Its FDP is correspondingly violated in the reference-cohort arms.

The positive control on ADHD-200 also reproduces the dependence on the number of true signals (1.00 at $k = 12$ against 0.24–0.30 at $k = 3$, at conditional $|t| = 6$), so the third cohort does not distinguish the mechanisms we identify.

The cohort-shift boundary is confirmed prospectively. The third cohort does provide an out-of-sample test of §3.6. ADHD-200 has nine sites whose protocols differ substantially, and its reference cohort is smaller by a factor of 3.2 ($n_{\text{ref}} = 121$ against 393). Its measured main/reference mismatch is 0.278, far beyond the simulated violation boundary of 0.12–0.19 (Figure 10b places it against the sampling-noise floors of all three cohorts), and its FDP behaviour follows: on the injected control the reference-cohort arms exceed the nominal level (FDP 0.22–0.29 across configurations), whereas the arm whose $\hat{\Sigma}$ comes from the main cohort itself—which is not a valid model-X procedure, but has no mismatch to get wrong—stays inside it (0.16–0.18). The contrast is what localises the failure to the mismatch between the two distributions, rather than to something particular about ADHD-200. In both ABIDE cohorts, whose mismatch is inside the safe region, the reference arms were controlled.

One caution is required, and it is the same caution that applies to any such comparison. A mismatch measured between two sample correlation matrices contains sampling noise as well as any genuine distributional difference, and tiny reference cohorts inflate it mechanically. We therefore measured the noise floor directly, by randomly splitting each cohort into two groups of the same sizes—so that the two groups differ only by sampling—and repeated this twelve times. On that scale ABIDE I and ABIDE II sit at 1.2× and 1.1× their noise floors, so their measured mismatch is almost entirely sampling; ADHD-200 sits at 1.7×, and the excess that remains after its own floor is subtracted, 0.117, lands at the lower edge of the simulated boundary. Figure 10b plots the observed mismatch against the floor, not this difference. The distinction matters because the two sources have different consequences—a biased $\hat{\Sigma}$ is anti-conservative, whereas a merely noisy one is conservative (Candès et al., 2018)—so a practitioner comparing mismatches should subtract the noise floor before comparing against the boundary.

4 Discussion

The standard account of reference-cohort knockoffs in neuroimaging is that the estimated covariance is too degenerate to build useful knockoffs from. We set out to quantify that account and found that it survives as a description of the spectrum but fails as an explanation of the outcome. The degeneracy is real; it costs about fourteen percentage points of power; it does not make the procedure return nothing. We then tested the obvious alternatives—six constructions, a method that never inverts Σ, a cross-environment consistency filter, and a continuous family of reparameterisations—and all of them fail at the same place. What remains is the feature geometry, and there the evidence is unambiguous: the conventional ROI summary is a row mean of a correlation matrix, hence a proxy for its first principal component, and that is what determines everything downstream.

The dominant choice is the one made first. Our meanconn features have effective rank 2.56 out of 113 because a row mean of a correlation matrix is a proxy for its first principal component (median R^2 of a single feature on PC1: 0.677). Partial correlation raises that to 71.6, and the increase is not an artefact of regularisation: with no shrinkage at all the effective rank is 59.8, and across shrinkage strengths from 0 to 0.5 it never falls below 53. The four standard summaries agree at median $r \leq 0.26$ per ROI, every ROI below 0.5—less agreement, on the same subjects and the same atlas, than the two preprocessing pipelines. A literature that reports which parcellation, which motion threshold and which statistical method was used, but not which ROI summary, is omitting the choice that moves the features furthest.

The reference cohort is a secondary constraint. Three independent manipulations agree that reference-cohort size is not what decides whether an analysis works, though it is not free either, as a comparison across arm strengths rather than a single row makes clear. Using the real covariance as ground truth gives power 0.859. Reference-cohort size from 100 subjects to the full 393 buys at most nine points of per-signal power, and only at moderate signal: at conditional $|t| = 3$ and $k = 12$ the increase is 0.590 to 0.676, narrowing to five points at $|t| = 6$, and the sign is the one theory expects from a noisier $\hat{\Sigma}$ (Candès et al., 2018). Cohort distribution shift at the level actually observed between ABIDE's cohorts *raises* power. And a method built to avoid Σ^{-1} entirely fails at the same place, for the same reason. Against the effect of the number of true signals—0.97 at $k = 12$ against 0.20 at $k = 3$—these are second order. This is a negative result about a widely assumed bottleneck, and its practical consequence is one of degree: enlarging a reference cohort buys a few points of power, while measuring whether the features are usable decides whether the analysis can work at all.

Two failure modes survive, and both are silent. Cohort shift destroys FDP calibration while inflating apparent power (§3.6), which is the direction that looks like success. The penalty constant of the feature statistic must satisfy a power condition and a solver-convergence condition simultaneously, and the software reports neither (§3.7). Both belong to a family that the matrix-power experiment makes vivid: operations that increase apparent signal while degrading calibration, whose effect on the numbers a user actually sees is an improvement. The general lesson is not specific to knockoffs. Any procedure whose error control is verified through an internally computed estimate needs that estimate to be validated under the manipulations the user is free to perform; the aggregating failure of §3.9 is the same lesson in a harder form, where the procedure began reporting discoveries that did not exist.

What a practitioner should do.

1. **Report the ROI summary's spectrum before any selection.** Effective rank, $\lambda_{\min}$ and the first-component share are computable from the reference cohort alone, with no pilot and no knockoff draw. On our data the conventional summary gives 2.56 of 113 with PC1 at

62%; partial correlation gives 71.6. A low effective rank means the features are close to one number, and no downstream method we tested rescues that.

2. **Report which ROI summary was used, and consider more than one.** The four standard summaries are almost mutually uncorrelated per ROI. Where the summaries disagree, the disagreement is the finding.

3. **Do not assume the conventional neuroimaging remedy applies.** Global-signal regression helped one pipeline, did nothing for another, and reduced the effective rank on ABIDE II. Removing the leading principal components worked everywhere we tried it.

4. **Do not expect a larger reference cohort to rescue the analysis.** It buys a few points of per-signal power at moderate signal (0.590 at 100 reference subjects against 0.676 at 393, both at $|t| = 3$, $k = 12$) and fewer at strong signal; that is worth having, but it is an order of magnitude smaller than the effect of how many coordinates are truly non-null.

5. **Report selections as distributions over draws, never as a single run.** In our data 35.5% of draws return nonempty selections, and a single draw would have justified the opposite conclusion.

6. **Validate the penalty constant on both axes before interpreting anything.** Check that a signal of known strength survives it and that the solver converges; report the convergence rate. The value we used originally failed the first test silently.

7. **Aggregate with e-values, never by averaging the statistics.** Averaging the W-vectors and thresholding manufactures discoveries under a global null.

8. **Check that cohort mismatch is inside the safe region.** Measure the RMS difference between the reference and main cohort correlation matrices. In our simulation FDP control is lost somewhere between mismatch 0.12 and 0.19; the observed ABIDE I value is 0.0593.

Relation to prior work. Our findings are consistent with the existing robustness theory and give it an empirical boundary. Barber et al. (2020) bound the FDR inflation under a misspecified F_X by a divergence measure; §3.6 traces that bound empirically and locates where it is crossed. Huang and Janson (2020) establish that exact control survives an estimated parametric family but are explicit that the choice of tuning parameters, including s, and the resulting power–robustness tradeoff are open; our results suggest the more consequential tuning parameter in practice is not s but the feature statistic's penalty. The construction literature's motivating pathology—reconstructability minimisation addressing the rank deficiency of the standard SDP solution when $\lambda_{\min}(\Sigma) \leq 0.5$ (Spector and Janson, 2022)—is a property of *knowing* Σ: estimating it by any shrinkage estimator, from any sample size, moves s off the constraint boundary. Here the same rank deficiency reappears from the opposite direction, as a ceiling on what any construction can achieve when the features themselves are nearly collinear. On the validity side, our cohort-shift result is the imaging analogue of reference-panel mismatch in knockoff GWAS (Chen et al., 2021;

Kanai et al., 2022), a problem for which no correction currently exists in that literature either.

Limitations. Seven should be stated plainly.

First, both cohorts study autism. ABIDE I and II are large and well-preprocessed but heavily analysed resources, and we make no claim about autism; the regions the procedure reports should be treated as a demonstration of method behaviour. Whether these findings extend to other disorders, to non-clinical phenotypes, or to non-imaging high-dimensional covariates is untested. We did add a third cohort of a different disorder (§3.11, ADHD-200), and it replicates the *ordering* of the summary definitions while showing that their severity is cohort-specific; but three cohorts, two of them autism, are not a broad sample, and the effective ranks we report for the conventional summary span a factor of 4.7 across them. The diagnostic is what we claim generalises, not any number in our tables. A fourth cohort—ideally a non-clinical or non-imaging one—would test that claim further.

Second, and most importantly, the discovery *sets* are not validated at any level. On real data we have no ground truth, so we cannot compute a false discovery proportion; the only error numbers in this paper come from injected signals and global nulls, where the truth is constructed. We therefore make no claim that any reported ROI is genuinely associated with diagnosis. We emphasise this because the failure modes we document are exactly the ones that produce plausible-looking output, and a positive result obtained under one configuration should not be assumed to survive another.

Third, the effective-rank severity is partly a property of our summary definition rather than of functional connectomes in general. That is the paper's subject, but the constant is not portable: a reader whose ROI summaries are less globally loaded should expect a larger effective rank and a smaller effect. We have not tested alternative ROI summaries such as system segregation, tangent-space covariance, or edge-level features, all of which are plausible alternatives and at least one of which (Faskowitz et al., 2020; Jo et al., 2021) has been shown to outperform nodal summaries for individual identification. Graphical-lasso precision is preferred to shrinkage-based partial correlation by the largest benchmark in this area (Peterson et al., 2023); we used the latter and report the unregularised effective rank as a check, but we did not compare the two directly.

Fourth, our simulations use Gaussian covariates and a linear response. The mechanisms we identify are properties of the construction, the covariance estimate and the feature geometry, but their quantitative constants may differ for other designs. The real response is binary and the statistic is a linear one; a logistic or nonparametric statistic may behave differently, though nothing in our argument depends on the statistic's functional form.

Fifth, the multi-environment comparison is underpowered in the configuration we ran. Its own positive control recovers only 0.100 of a strong injected signal, so its empty selection on the real response carries no information about the data; with larger environments the same control recovers 0.762 (§3.4). A version of that test with site-level samples large enough to have power—by pooling sites or by using a cohort with fewer, larger sites—remains to be run, and would be the natural way to make the cross-site question answerable.

Sixth, one alternative statistic could not be evaluated. Masked likelihood-ratio statistics (Spector and Fithian, 2024) are penalty-free and asymptotically optimal and would remove the tuning problem of §3.7 entirely; our implementation did not satisfy the statistic's defining null-symmetry property and we report no results from it. Establishing whether it succeeds where the statistics we tested do not is a natural next step.

Seventh, we do not provide a formal theory for the divergence between reported and true FDP, nor for the FDP boundary in cohort mismatch, nor for the dependence of power on the number of non-nulls in this regime. We measure all three and describe their behaviour; theoretical treatments would be valuable.

Conclusion. The reference cohort is not what limits these analyses, and neither, in the range we can measure, is the conditioning of the covariance: across 41 configurations spanning effective rank 2.2 to 113, per-signal power never falls below 0.77. What the ROI summary statistic decides is the geometry of the feature space, and that choice—made by convention, measurable before any analysis is run, and not currently reported—moves the features further than any downstream decision we could find. What limits power is the number of true signals; what limits validity is cohort mismatch and the penalty constant. We recommend reporting all four, and reporting every selection as a distribution over draws rather than as a single answer.

Declarations

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Competing interests. The authors have no relevant financial or non-financial interests to disclose.

Use of large language models. An AI coding assistant (Claude, Anthropic) was used to draft and debug analysis code, to run and tabulate the experiments reported here, and to assist with manuscript preparation. All experimental designs, all scientific claims, and every reported number were directed, checked and approved by the authors, who take full responsibility for the content.

Ethics approval. Not applicable. This study is a secondary analysis of three publicly available, de-identified datasets (ABIDE I, ABIDE II and ADHD-200). Ethical approval and informed consent were obtained by the original data-collecting studies and by the data-sharing initiatives distributing them; no new data were collected, and no participant-identifiable information was accessed.

Consent to participate / Consent for publication. Not applicable (secondary analysis of de-identified public data).

Availability of data and material. ABIDE I was obtained from the FCP-INDI public archive (`https://s3.amazonaws.com/fcp-indi`), branches `cpac/filt_noglobal`

and `niak/filt_noglobal`; these require neither registration nor a data use agreement. ABIDE II was obtained from the same archive (`Derivatives/DCAN`, no registration required) for the imaging, and from the NITRC Neuro Bureau file release area (`https://www.nitrc.org/frs/?group_id=383`) for the composite phenotypic file, which requires a NITRC account and agreement to the 1000 Functional Connectomes Project terms. ADHD-200 (The ADHD-200 Consortium, 2012) was obtained from the Neuro Bureau ADHD-200 Preprocessed repository (Bellec et al., 2017) (`http://fcon_1000.projects.nitrc.org/indi/adhd200/`, no registration required), which distributes the Athena-preprocessed images and their quality flags. Derived feature matrices for all three cohorts are deposited at [repository DOI, to be assigned].

Code availability. All analysis code—feature construction for all four ROI summaries, the knockoff pipeline, the boundary, positive-control, cohort-shift, draw-distribution, construction-comparison, Gaussian-mirror, multi-environment and matrix-power experiments, and the scripts that generate every figure and table—is available at `[repositoryDOI,tobeassigned]`. Every figure is accompanied by the exact numbers it draws, as `paper/figs/source_data/<figure>.csv`, alongside the 300 dpi raster and vector versions of the figure itself. The archive includes a provenance record for every result file and a `verify_claims.py` script that registers every numeric claim together with the result file and cell that supports it, checks each one, and fails loudly if any cannot be traced. The same script scans the manuscript for numeric literals that no claim covers.

Authors' contributions. J.L. conceived the study, designed the experiments and implemented the analysis; S.Z. and X.J. contributed to the experimental design and to the interpretation of the results. J.L. wrote the manuscript with input from all authors. All authors read and approved the final manuscript.

Information Sharing Statement. All analysis code, the derived feature matrices and the complete result tables are openly available; the archive is referenced under Code availability above and contains a script that registers every numeric claim with the specific result file and cell that supports it, and checks them all. Raw imaging data are not redistributed here, but both source archives are public and are identified, with their access conditions, in Availability of data and material.

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